Danube Indeet Model – User Manual

Version 0.3.10

Decision support tool for modelling infrastructure of combined energy and transport hubs.

May 6, 2026

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Preface

This is the documentation for version 0.3.10. (Version history). For download link and other useful information see the main model website.

This user manual introduces the **Danube Indeet Model (DIM)**, a practical and user-friendly solution for planning and sizing investments in integrated energy and transport hubs based on renewable energy. The tool enables users to evaluate how locally available or imported renewable energy can be optimally used to charge EVs and/or to be converted into green hydrogen, while also efficiently integrated with electricity, gas, heating, and water networks.

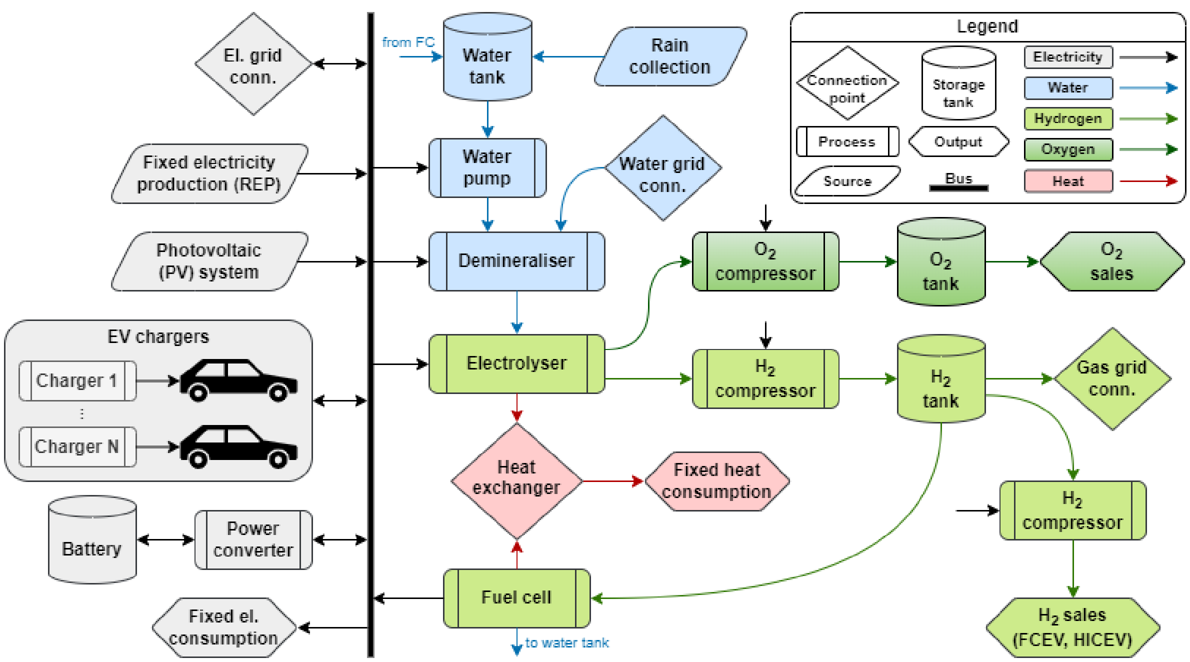

Figure 1: Diagram of the overall system configuration that can be modelled.

The DIM delivers techno-economic insights, including optimal system configuration, CAPEX/OPEX estimates, and key financial indicators such as NPV, IRR, and ROI. Designed for municipalities, energy planners, development agencies, and investors, the DIM supports informed, data-driven decisions for developing cost-effective and future-ready hydrogen hubs.

Key Features of the DIM:

- **Flexible EV charging including V2G** Smart charging of EVs, including vehicle-to-grid, helps to balance the grid.
- **Green hydrogen focus** Convert local and regional renewable energy surpluses into clean, versatile hydrogen.
- **Multi-grid integration** Seamlessly plan interactions across electricity, gas, heating, and water networks.
- **Flexible renewable energy input** Input renewable energy manually, or define local solar PV capacity.
- **Comprehensive system modelling** Understand every component and how they interact within the overall system.
- **Techno-economic analysis** Get CAPEX/OPEX estimates and financial indicators like NPV, IRR, ROI, and payoff period.
- **Optimal sizing & operation** Receive recommendations for sizing electrolysers, fuel cells, photovoltaics, wind farms, energy storage, and more, plus operation schedules.
- **Intuitive graphical interface** User-friendly GUI for easy input, visualization, and interpretation of results.
- **Built on proven optimization tools** Leverages and expands the DanuP-2-Gas Optimization Tool for advanced system planning.
- **Targeted user support** Designed for municipalities, energy planners, development agencies, and private investors.
- **Scenario analysis capability** Explore dynamic planning scenarios to optimize infrastructure, investment, and performance.

Acknowledgements

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The tool is developed and maintained by:

University of Zagreb Faculty of Electrical Engineering and Computing
Department for Control and Computer Engineering
Laboratory for Renewable Energy Systems
Zagreb, Croatia
lares@fer.hr

Further development

To report issues, request changes, ask questions, or provide feedback, email us to: filip.rukavina@fer.hr; ivan.grabic@fer.hr; marko.kovacevic@fer.hr; mario.vasak@fer.hr, with indication whether it is about:

- A **bug** – the tool does not operate as it should. Make sure to:
  - tell us what you expected to happen,
  - tell us what actually happened,

  - tell us what version of the DIM you use,
  - send us screenshots of the problem,
  - send us the project file (.hyefre) where it happens.
- A **change** – you would like to see an improvement, whether it is to change an existing feature or adding a new one. Make sure you explain it in your email and tell us which version of the tool you are currently using.
- **Questions** – just ask if you have questions or need assistance with tool usage.
- **Documentation** – let us know if something is wrong or missing in the documentation.

License

How to cite

TODO: add article to cite

Version history

- feature: added user manual

2026-04-23 v0.3.9 build 009

- bugfix: BESS SOE now multiplied by timestep duration to track energy
- bugfix: freeze GUI fix when optimizing without saving
- feature: changed message for negative recommendations
- feature: don't show recommendation if N negative
- bugfix: EV OPEX calculations changed
- feature: Run all optimizations added in Tools
- bugfix: PV profile PDF export
- bugfix: cost of REP in results
- bugfix: waiting screen locked when optimizing without saving

2025-12-17 v0.3 build 008

- feature: export results to PDF
- bugfix: energy export with REP
- bugfix: selling H2 to gas grid
- bugifx: values of default parameters
- bugfix: upper bounds in LP
- bugfix: weekly and monthly sales
- other fixes

2025-11-28 v0.3 build 007

- results showing full data

2025-11-27 v0.3 build 006

- bugfixes for results

2025-11-27 v0.3 build 005

- feature: optimization is running
- feature: results of the optimization are displayed
- feature: advanced parameters of the electrical grid connection
- bugfixes

2025-10-21 v0.2 build 004

- refactor: project has only .indeet file (no more folders)
- refactor: all profiles are contained in the .indeet file
- refactor: temp and log folder are moved to application root folder
- refactor: logs are still project-based, one file for each created project
- bugfix: project files can be opened on any computer
- new feature: validation of input parameters
- bugfix: default simulation start and end dates corrected
- bugfix: electrolyser and fuel cell parameters calculation corrected
- bugfix: predefined parameters of electrolyser and fuel cell updated and corrected

2025-08-26 v0.2 build 003

- fixed reading images from non-ASCII paths

2025-07-28 v0.2 build 002

- initial working GUI